



UNIVERSITÀ DEGLI STUDI DI PADOVA

STOCHASTIC METHODS FOR FINANCE

Report 8: Pricing of European Options under Stochastic Volatility – the Heston Model

Proff.

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1 Introduction

This report documents the comprehensive implementation of the Heston Stochastic Volatility Model. By contrasting a Full Truncation Monte Carlo scheme with a semi-analytical Fourier Transform approach, this study evaluates the model's robustness and pricing accuracy, particularly in the presence of Feller condition violations and varying strike price regimes.

2 Heston Model

The dynamics of the asset price S_t and the variance process v_t are described by the following coupled stochastic differential equations:

$$\begin{aligned}dS_t &= rS_t dt + \sqrt{v_t}S_t dW_t^S \\ dv_t &= \kappa(\theta - v_t)dt + \xi\sqrt{v_t}dW_t^v\end{aligned}$$

where the two Brownian motions are correlated according to $\text{Corr}(dW_t^S, dW_t^v) = \rho$. The model parameters have the following interpretation:

Symbol	Interpretation
r	Risk-free interest rate
v_t	Instantaneous variance of the asset
κ	Speed of mean reversion of the variance process
θ	Long-term average variance
ξ	Volatility of volatility ("vol of vol")
ρ	Correlation between asset returns and variance shocks
W_t^S, W_t^v	Correlated Wiener processes

2.1 Numerical Implementation and Truncation Scheme

Under standard Euler-Maruyama discretization, the variance process in the Heston model can occasionally take negative values due to numerical error. This risk is significantly heightened when the Feller condition, defined as $2\kappa\theta \geq \xi^2$, is violated. Because the diffusion terms for both the asset price and the variance involve $\sqrt{v_t}$, negative values result in invalid numerical operations, leading to simulation failure and NaN outputs.

To ensure the stability and positivity of the simulated paths, I implemented a "Full Truncation" scheme. In this approach, the variance used in the drift and diffusion coefficients is replaced by $\max(v_t, 0)$. The discretized equations used in the Python implementation are:

$$\begin{aligned}S_{t+\Delta t} &= S_t + rS_t\Delta t + \sqrt{\max(v_t, 0)}S_t\Delta W_t^S \\ v_{t+\Delta t} &= v_t + \kappa(\theta - \max(v_t, 0))\Delta t + \xi\sqrt{\max(v_t, 0)}\Delta W_t^v\end{aligned}$$

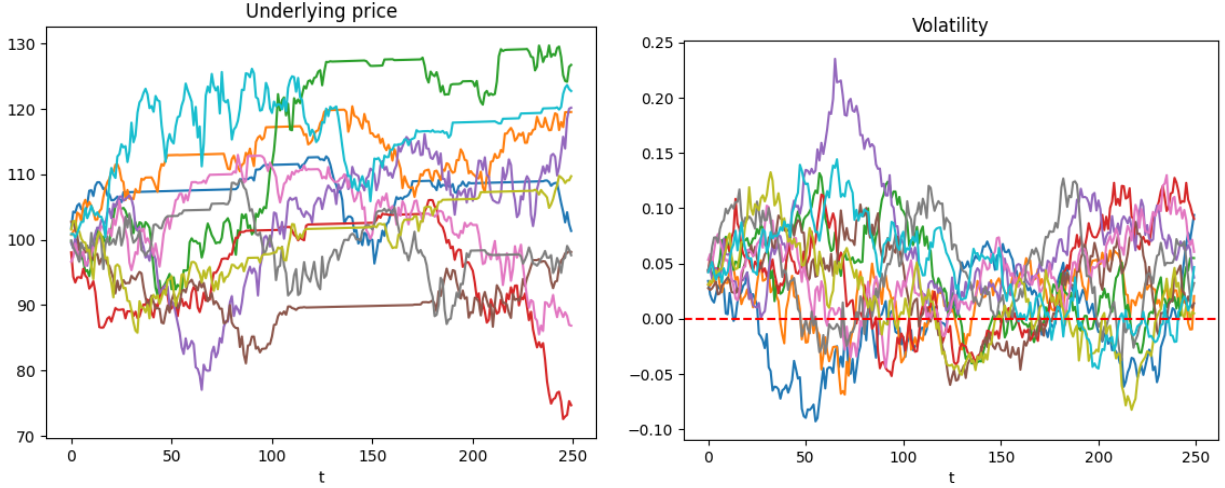


Figure 1: Simulated underlying and volatility paths

As shown in Figure 1, the truncation scheme effectively prevents numerical instabilities, allowing all simulated paths to remain finite. However, a qualitative artifact of this method is visible: when the variance is truncated to zero, the stochastic diffusion term disappears temporarily. This produces locally smooth or almost linear trajectories in the asset price until the variance reverts above zero.

3 Terminal Distribution Analysis: Heston vs. Black-Scholes

To assess the impact of stochastic volatility, a Black-Scholes (BS) benchmark was simulated using a constant volatility $\sigma = \sqrt{\theta}$, ensuring the average variance matches the long-term mean of the Heston process. Analytically, the terminal asset price in the Black-Scholes model follows a lognormal distribution:

$$\ln(S_T) \sim \mathcal{N}\left(\ln(S_0) + \left(r - \frac{\sigma^2}{2}\right)T, \sigma^2 T\right)$$

3.1 Moments and Convergence

The statistical moments of the terminal distributions are summarized in Table 1.

The proximity of the simulated means (101.75 for Heston and 102.00 for BS) to the analytical Black-Scholes value of 102.02 validates the convergence of the Monte Carlo simulation and confirms the correct implementation of the drift component.

The variance of the Heston terminal distribution (702.74) is significantly higher than that of the Black-Scholes model (422.85). While volatility is a "locked" parameter in Black-Scholes, the randomness of the Heston variance process creates a "spreading" effect on terminal prices.

The Black-Scholes model exhibits a positive skewness (0.58), inherent to the lognormal distribution. In contrast, the Heston model produces a negative skewness (-0.28), capturing the "leverage effect" where price drops are correlated with higher volatility, resulting in fatter left tails and higher downside risk.

Both models are leptokurtic, showing positive excess kurtosis. While Black-Scholes kurtosis is a fixed property of its lognormal shape, the Heston kurtosis (0.58) is driven by the volatility-of-volatility (ξ), allowing it to model a higher frequency of extreme price movements.

Statistic	Heston	BS Sim.
Mean	101.75	102.00
Variance	702.74	422.85
Skewness	-0.28	0.58
Kurtosis	0.58	0.51

Table 1: Comparative statistics.

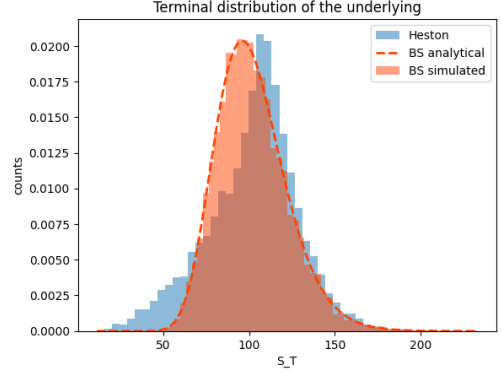


Figure 2: Terminal distribution: Heston vs. BS.

4 Option Pricing Comparison: Heston vs. Black-Scholes

In this section, the European Call option prices generated by the Heston Monte Carlo simulation are compared against the analytical Black-Scholes prices..

The following table summarizes the premiums obtained for various strike prices K :

Strike (K)	Heston Call Price	BS Call Price
90	17.65 ± 0.18	14.81
100	11.19 ± 0.15	8.92
110	6.27 ± 0.12	4.94

The Heston model consistently prices the Call options higher than the Black-Scholes model across all tested strikes. This systematic difference indicates that the pricing premium in the Heston framework is primarily driven by the wider dispersion of the terminal asset prices.

The stochastic nature of the variance increases the overall variance of the terminal distribution, which in turn increases the expected value of the convex payoff function $\max(S_T - K, 0)$. This leads to higher premiums even when the "fat-tailedness" (kurtosis) of the distribution remains comparable to the lognormal case.

5 Implied Volatility and the Volatility Skew

To further analyze the characteristics of the Heston model, I computed Call option prices for a wide range of strikes ($K \in [70, 130]$) and subsequently calculated the Black-Scholes implied volatility (IV) using a bisection root-finding algorithm.

5.1 Analysis of the Volatility Skew

The resulting implied volatility surface is presented in Figure 3.

- **Volatility Skew:** The downward-sloping implied volatility curve confirms the presence of a volatility skew. This is a direct consequence of the negative correlation ($\rho = -0.7$) between the asset price and its variance, effectively modeling the leverage effect where market drops are associated with higher volatility.
- **Volatility Premium:** The implied volatility remains consistently above the 20% ($\sqrt{\theta}$) baseline across all strikes. This illustrates the "volatility premium" demanded by the Heston model to account for the higher total variance generated by the stochastic volatility process compared to the constant volatility Black-Scholes framework.

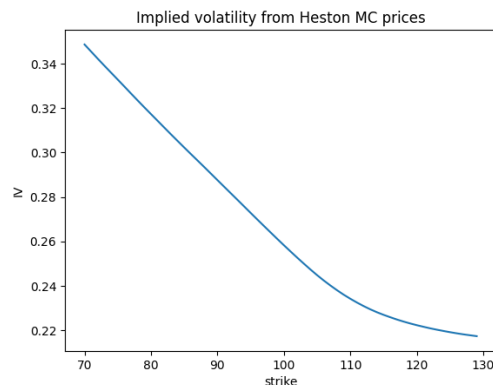


Figure 3: Implied volatility curve derived from Heston Monte Carlo prices.

5.2 Sensitivity Analysis: Impact of Correlation (ρ) and Volatility of Volatility (ξ)

In this task, the sensitivity of the implied volatility (IV) surface to the correlation parameter ρ and the volatility-of-volatility parameter ξ is examined. These parameters allow the Heston model to capture the volatility skew and smile—phenomena that the constant-volatility Black-Scholes model cannot replicate.

5.2.1 Impact of Correlation (ρ): The Leverage Effect

As illustrated in Figure 4, the correlation parameter governs the slope and asymmetry of the IV curve:

- **Leverage Effect ($\rho < 0$):** As the correlation becomes more negative (e.g., moving from -0.5 to -0.9), the curve steepens. This reflects the market reality where price drops trigger higher volatility, making downside protection at lower strikes more expensive.

- **Volatility Smile ($\rho = 0$):** In the absence of correlation, the curve transforms into a symmetric "smile." This shape is driven by the kurtosis introduced by the stochastic variance, which increases premiums for extreme price movements in either direction.

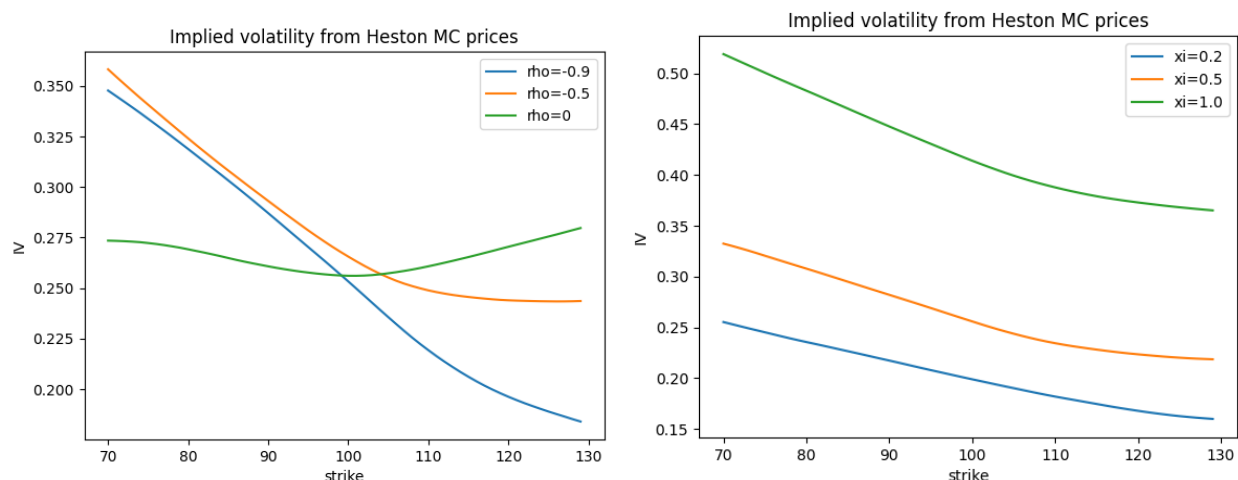


Figure 4: Sensitivity of implied volatility to parameters ρ and ξ .

5.2.2 Impact of Vol-of-Vol (ξ): Tail Fatness and Magnitude

Figure 4 demonstrates how the ξ parameter controls the magnitude and curvature of the IV surface:

- **Vertical Shift:** Increasing ξ from 0.2 to 1.0 causes a significant upward shift across the entire curve. This represents the "volatility premium," where added uncertainty in the variance process increases option premiums across all strikes.
- **Impact on Kurtosis:** Higher vol-of-vol leads to "fatter tails" (higher kurtosis) in the terminal distribution. This effect dramatically raises the overall level and convexity of the IV curves.
- **Smile Accentuation:** While the general slope is maintained by ρ , a higher ξ accentuates the "smile" component. This is most visible in the $\xi = 1.0$ case (green line), which shows a sharper reaction at extreme strikes compared to the flatter $\xi = 0.2$ case (blue line).

6 Semi-Analytical Pricing: Fourier Transform Method

The final task involves the implementation of the Heston characteristic function to price European Call options via numerical integration. This provides a semi-analytical benchmark to validate the Monte Carlo results.

6.1 Heston Characteristic Function

The characteristic function of the log-price X_T under the Heston model is a closed-form solution to the associated partial differential equation. To ensure numerical stability and avoid branch-cut issues (the "Heston Trap"), the formulation is implemented as:

$$\phi(z; X_0, v_0, \tau) = \exp(izX_0 + A(\tau, z) + B(\tau, z)v_0)$$

Where the coefficients $A(\tau, z)$ and $B(\tau, z)$ are defined by:

$$B(\tau, z) = \frac{-(iz + z^2)(1 - e^{-\delta\tau})}{\psi_+ - \psi_- e^{-\delta\tau}}$$

$$A(\tau, z) = riz\tau - \frac{\kappa\theta}{\xi^2} \left[2 \ln \left(\frac{\psi_+ - \psi_- e^{-\delta\tau}}{2\delta} \right) + (\delta - \psi_+)\tau \right]$$

With the auxiliary variables:

$$\delta = \sqrt{(\kappa - i\rho\xi z)^2 + \xi^2(iz + z^2)}$$

$$\psi_+ = \kappa - i\rho\xi z + \delta$$

$$\psi_- = \kappa - i\rho\xi z - \delta$$

The price of an option with a terminal payoff $g(X_T)$ is calculated as:

$$\text{Price}_t = \frac{e^{-r(T-t)}}{2\pi} \int_{-\infty - i\eta}^{\infty - i\eta} \phi(z) \hat{g}(-z) dz$$

For a Call option with strike K , the Fourier transform of the payoff function is given by:

$$\hat{g}(z) = -\frac{K^{iz+1}}{z^2 - iz}$$

The integration is performed along a complex line $z = u - i\eta$ with a dampening factor $\eta = 1.5$. This choice ensures that $\text{Im}(-z) > 1$, which is a necessary condition for the convergence of the Call payoff transform.

6.2 Comparison of Fourier vs. Monte Carlo Prices

The table below summarizes the convergence between the semi-analytical Fourier results and the Monte Carlo simulations ($M = 10,000$, $N = 250$).

Efficiency Analysis

- **Fourier Transform Method:** This approach is the most efficient for pricing European options as it reduces the problem to a one-dimensional numerical integration of the characteristic function. While Monte Carlo requires simulating thousands of paths, the Fourier-Simpson method calculates prices almost instantaneously.

Strike	$C_{Fourier}$	C_{MC}	Abs Error
80	23.00	23.67 ± 0.17	0.6717
90	15.08	15.92 ± 0.15	0.8388
100	8.67	9.42 ± 0.12	0.7480
110	4.18	4.62 ± 0.09	0.4470
120	1.61	1.91 ± 0.06	0.3069

Table 2: Validation of Monte Carlo prices against the Fourier benchmark.

- Monte Carlo Method: The computational cost is significantly higher due to the iterative nature of the Euler-Maruyama discretization. Generating $M = 10,000$ paths across $N = 250$ time steps involves substantial memory and processing time, making it less ideal for high-frequency pricing of standard European derivatives.

Accuracy and Convergence

- Analytical Benchmark: The Fourier transform results represent the semi-analytical "exact" prices for the Heston model.
- Numerical Validation: The small Absolute Error (ranging from 0.31 to 0.84) validates the Full Truncation Monte Carlo scheme. It proves that the simulation correctly tracks the analytical benchmark, with the remaining discrepancy attributed to discretization bias and sampling noise.
- Feller Condition Resilience: Although the Feller condition ($2\kappa\theta \geq \xi^2$) is violated ($0.16 < 0.25$), the results remain robust. The Monte Carlo scheme successfully manages the variance process hitting zero without losing accuracy, while the Fourier method remains mathematically stable regardless of the Feller status.

For European options, the Fourier method is the superior tool for speed and precision. However, the close convergence between the two methods confirms that your Monte Carlo implementation is highly reliable for more complex valuation tasks where analytical solutions are unavailable.